Qian Qian

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Skills

- **Languages**: Java, Kotlin, Python, Rust, C, C++, Shell, Swift, Go, Typst, LaTeX
- **Tech/Tools**: Git, GitHub Actions, Linux, Unit Testing, JUnit, PyTest, NGINX, Docker, Ktor, Flask, SwiftUI, Quilt

Experience

Software Developer & Software Subsystem Co-Lead September 2025 — Present
Waterloo Rocketry *Waterloo, Ontario, Canada*

- Built Mission Control deployment infrastructure to ensure consistent and reproducible software environments using Docker containerization and Ansible configuration management.
- Developed Python-based data acquisition and processing systems to improve reliability of Mission Control telemetry workflows, supported by PyTest unit testing.
- Assembled and tested ground-side electrical and sensor systems to support reliable launch-pad operations during integration and pre-launch testing.
- Mentor developers to improve team-wide engineering practices through development standards, Git workflows, unit testing, pull-request reviews, and collaborative debugging.

Software Engineering Intern, Build and Automation May 2026 — August 2026
Maxon *Montreal, Quebec, Canada*

- Built a Redshift rendering performance-monitoring system to identify regressions across standard rendering unit tests using Python, Grafana, and Loki for data processing, aggregation, and visualization.
- Automated ArchVis changelog generation and build notifications to improve QA visibility using Python and GitLab and Jira APIs.

Projects

QubeC - *Rust, Zig, C3, C++, Go* August 2025 — Present

- Built a bytecode compiler and stack-based virtual machine for a custom C-like programming language.
- Explored multiple programming languages through compiler development and smaller experimental projects to compare language features, tooling, ergonomics, and performance characteristics.

Typstify - *Swift, SwiftUI, Rust* July 2024 — Present

- Developed a native Typst editor for iPadOS to support document authoring and preview workflows using SwiftUI with a Rust-based Typst backend, bridged via swift-bridge for cross-language interoperability.
- Contributed patches to 3 upstream dependencies to expand functionality and resolve existing issues.

Maintainer, ungoogled-chromium-macos - *Shell, Python, Build Systems* November 2023 — Present

- Maintained the macOS distribution of ungoogled-chromium, a privacy-focused browser, delivering 100+ releases to 100,000+ users through upstream tracking, build-script maintenance, and downstream patch management.
- Reduced CI build time by 40% by optimizing GitHub Actions workflows to accelerate build time and release cycles.

CalcLox - *Java* January 2025 — May 2025

- Designed and implemented a Lox-based scripting language and its interpreter for a programmable calculator.
- Built JUnit unit tests to validate language semantics and interpreter correctness across core calculator workflows.

Education

University of Waterloo Waterloo, Ontario, Canada
Candidate for Bachelor of Software Engineering, Honours Software Engineering, Co-op *September 2025 — May 2030*

Additional Information

- **Honour and Awards**: Winner of Hack the North 2025 QNX Software Challenge, Canadian Senior Mathematics Contest Distinction (2024), 2 x Canadian Computing Competition Distinction (2023, 2024)
- **Activities**: Programming Competition Problem Setter at CRC Robotics